

CD2 - AT2 - CLASS TEST

1. a) Define Data mining and list any two basic data mining task then explain KDD.

Answer:

DATAMINING:

Is the process of extracting useful patterns, relationship and knowledge from large amount of data using statistical and ML techniques.

TASK:

1. Classification : Assigns data into predefined categories
2. Clustering : Grouping similar data without predefined classes.

KDD:

1. Data cleaning
2. Data integration
3. Data selection
4. Data transformation
5. Data mining
6. Pattern Evaluation
7. Knowledge Presentation.

b) What is Data Cleaning ? & common issues, Explain handle missing value and noisy data.

Answer:

DATA CLEANING:

Is the process of detecting and

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Correcting inaccurate, duplicate data.

Two common Issues:

1. Missing value
2. Noisy data.

Handling Missing value / Noisy data.

- Ignore the record
- Fill with Mean, median & mode
- Use Previous or next value.
- Predict using ML.
- Binning
- Regression
- Clustering.

2. a) Define data integration, 2 challenges then Explain handling redundancy & inconsistency

Answer:

DATA INTEGRATION:

Is process of combining data from multiple source into single Dataset.

2 Issues:

1. Data redundancy
2. Data inconsistency.

Handling Issues:

- Remove duplicate record
- Use Primary key
- Standardize format
- Convert all unit to same measurement.

b) What is Co-variance Analysis?
List two formula.

Answer:

CO-VARIANCE ANALYSIS:

- Positive Co-variance $\rightarrow$ variable increases together
- Negative Co-variance $\rightarrow$ one increases while the other decreases
- Zero Co-variance $\rightarrow$ No relationship

Co-variance formula.

Population :
$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample :
$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

3. a) Given stocks:

A = 2, 3, 5, 4, 6

B = 5, 8, 10, 11, 14

As stock A generally increases, stock B also increases.

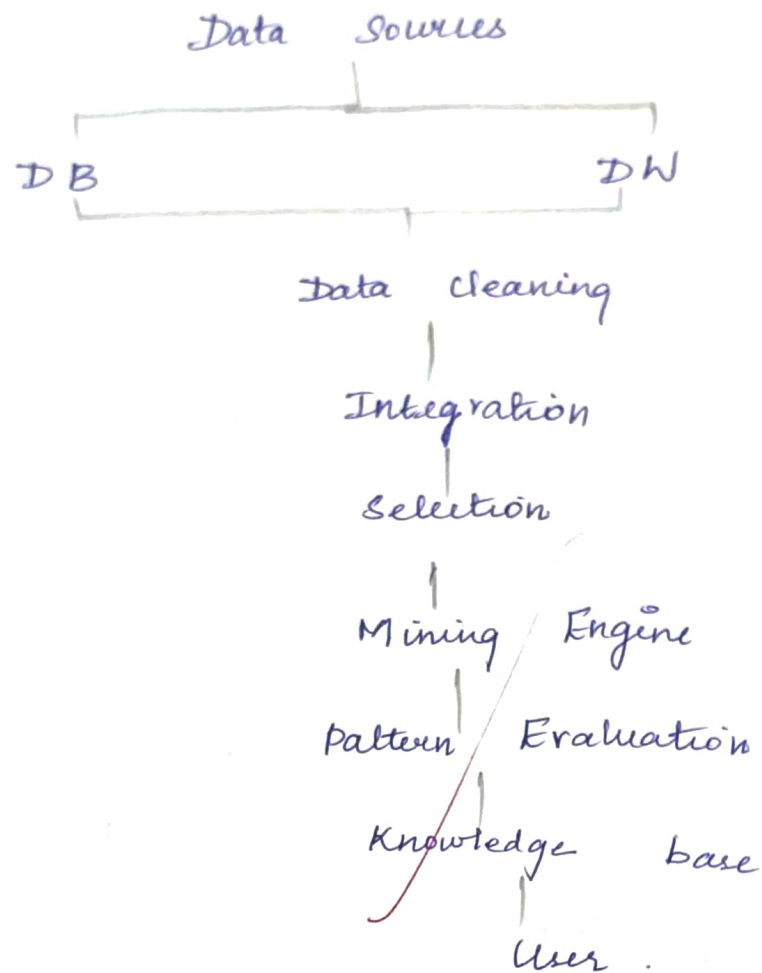
Co-variance is positive

$\therefore$ The Both stock prices rise together

They are not independent because they are affected by same industry trends.

b) Explain the Architecture of typical Data mining system.

DIAGRAM:



a) Find Co-variance.

Given:

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = \bar{x} = 6.2$$

$$\text{Mean}(Y) = \bar{y} = 5.8$$

x	y	(x - $\bar{x}$)	(y - $\bar{y}$)	product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

Sum = -1.8

$$\therefore \text{Covariance} \Rightarrow \frac{-1.8}{5-1} = -0.45$$

b) Describe major process in data preprocessing.

1. Data cleaning
2. Data integration
3. data Transformation
4. data reduction
5. data discretization

5) a) Perform Equal - Frequency .

Bin 1 : 13, 15, 16, 16, 19, 20, 20, 21, 22
 Mean : $162/9$
 = 18 .

Smoothed values : 18, 18, 18, 18, 18, 18, 18, 18, 18

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Mean: $253/9 = 28$.

Smoothed value: 28, 28, 28, 28, 28, 28, 28, 28, 28

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Mean: $394/9 = 44$

Smoothed value: 44, 44, 44, 44, 44, 44, 44, 44, 44

b) Explain Data Discretization. Compare Equal width & Equal frequency Binning.

Data Discretization:

Converts continuous numerical values into limited no of intervals.

Comparison:

Equal - width	Equal - frequency
Equal interval size	Equal no. of records
Simple to implement	Better handles data
may create empty bin	contain similar values
Sensitive to outliers	less affected by outliers.